

QST BA 780

Team 4

Global Semiconductor Supply Chains

Trade Flows, Risks, and Resilience

Agenda

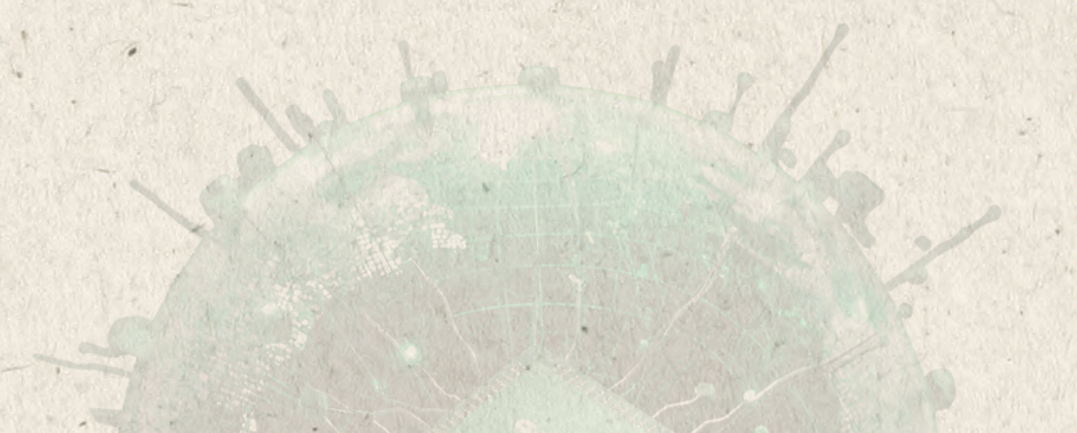
1. Problem Definition

2. Motivation

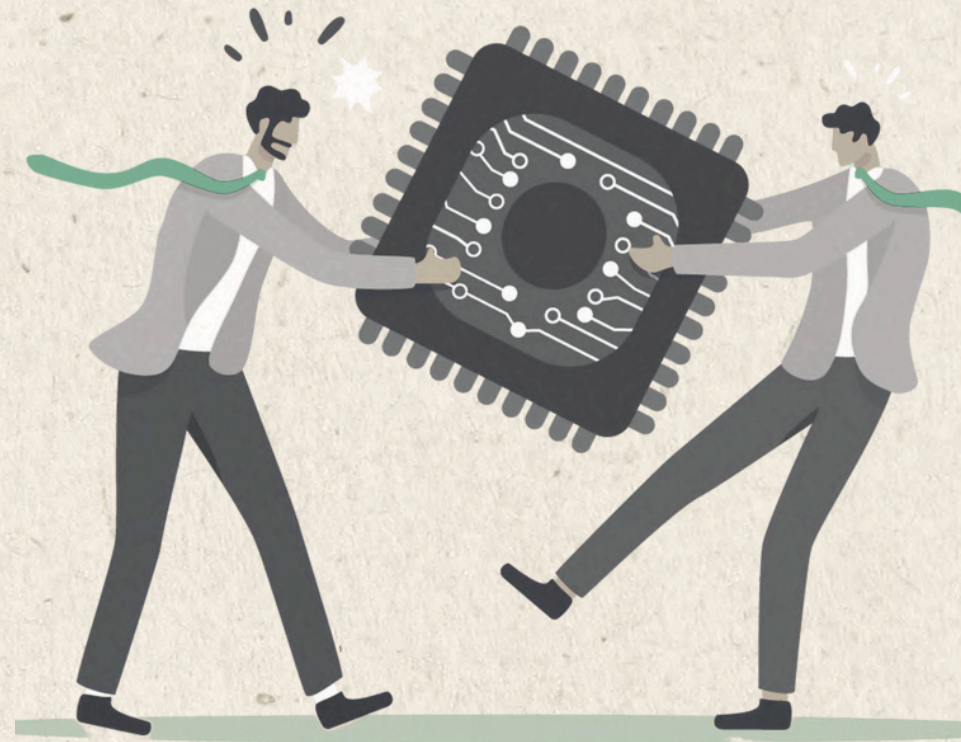
3. Vizualizations

4. Conclusion

Problem Definition

- Investigates how international semiconductor trade impacts supply chain concentration and operational risk.
 - Analyzes import/export dependencies and identifies regional chokepoints that affect global distribution.
 - Examines shifts in semiconductor flows over time using transaction-level trade data by country and commodity.
 - Connects trade patterns to operational disruptions and financial resilience in the industry.
 - Aims to provide actionable insights for mitigating future supply shortages and enhancing global resilience.
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Motivation



Semiconductors the infrastructure of modern life

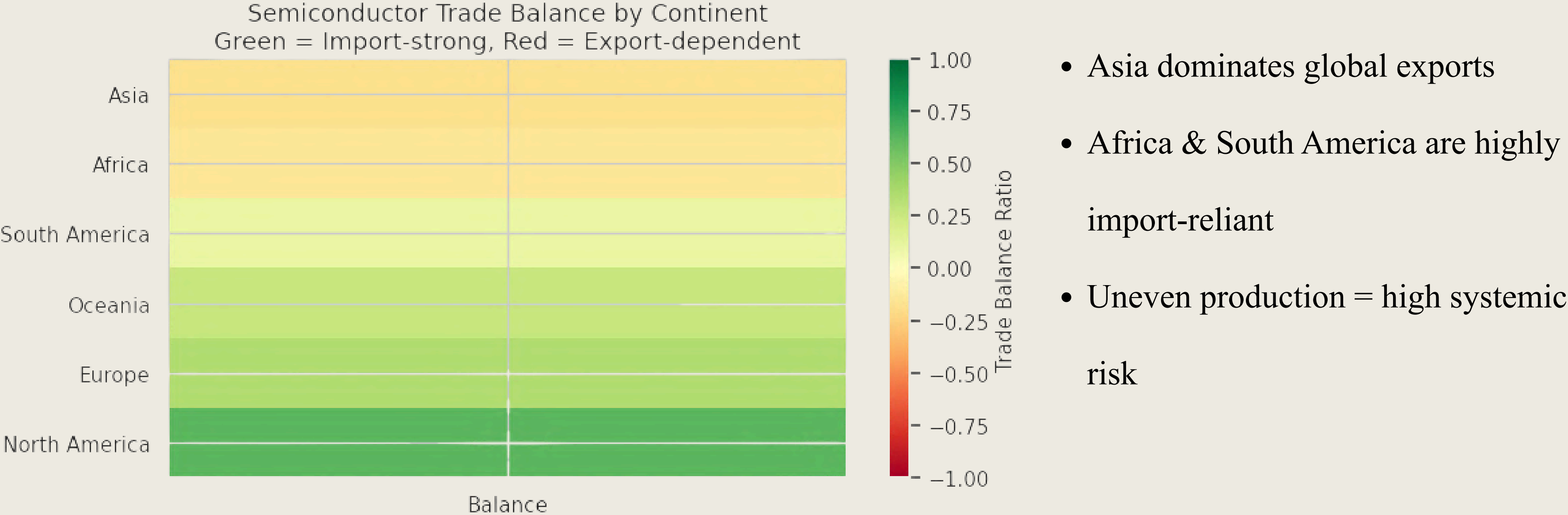
Import/export dependencies and chokepoints heighten operational risk.

Recent shortages (2019–2022) exposed how global dependence on few producers amplifies risk.

Data fragmentation prevents clear visibility into trade dependencies and chokepoints

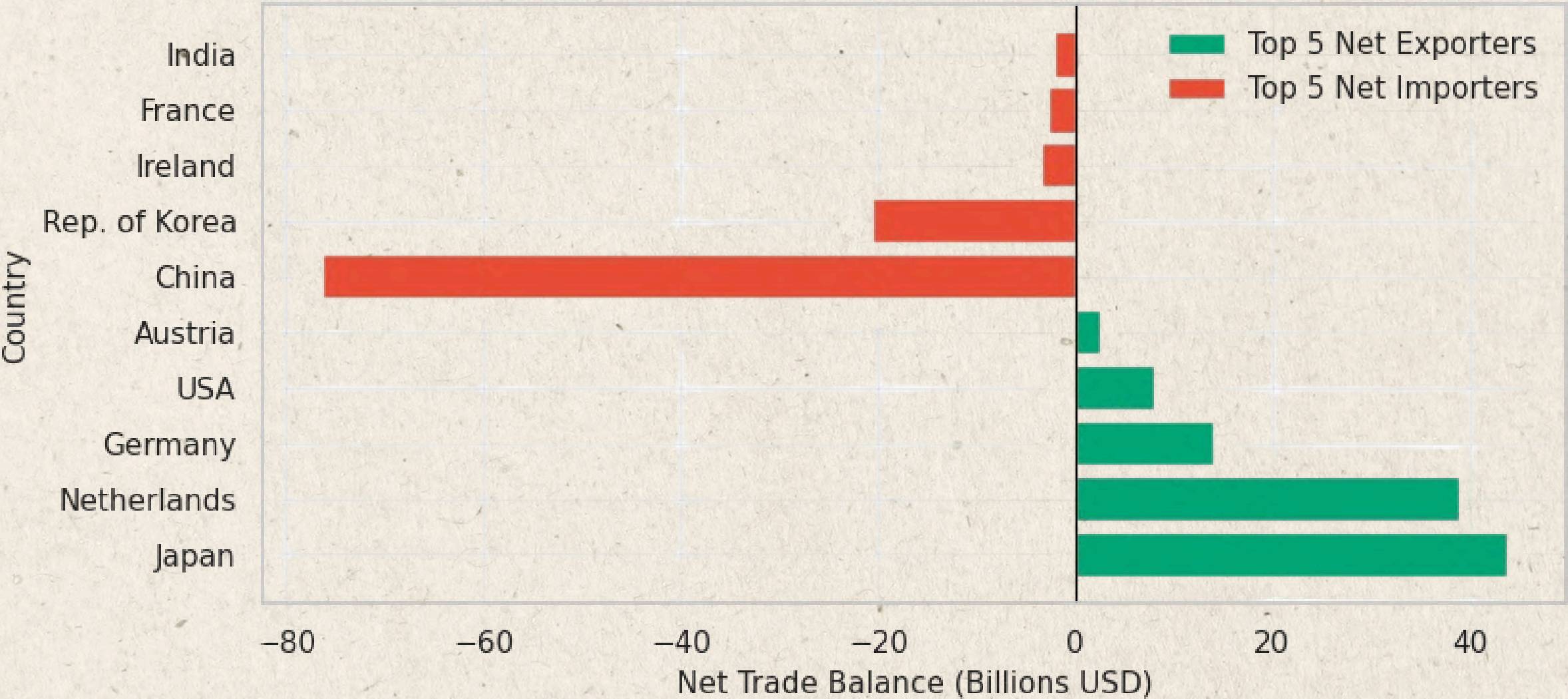
1. Global Trade Patterns and Shifting Power Dynamics

Global Trade Landscape



Global Trade Landscape

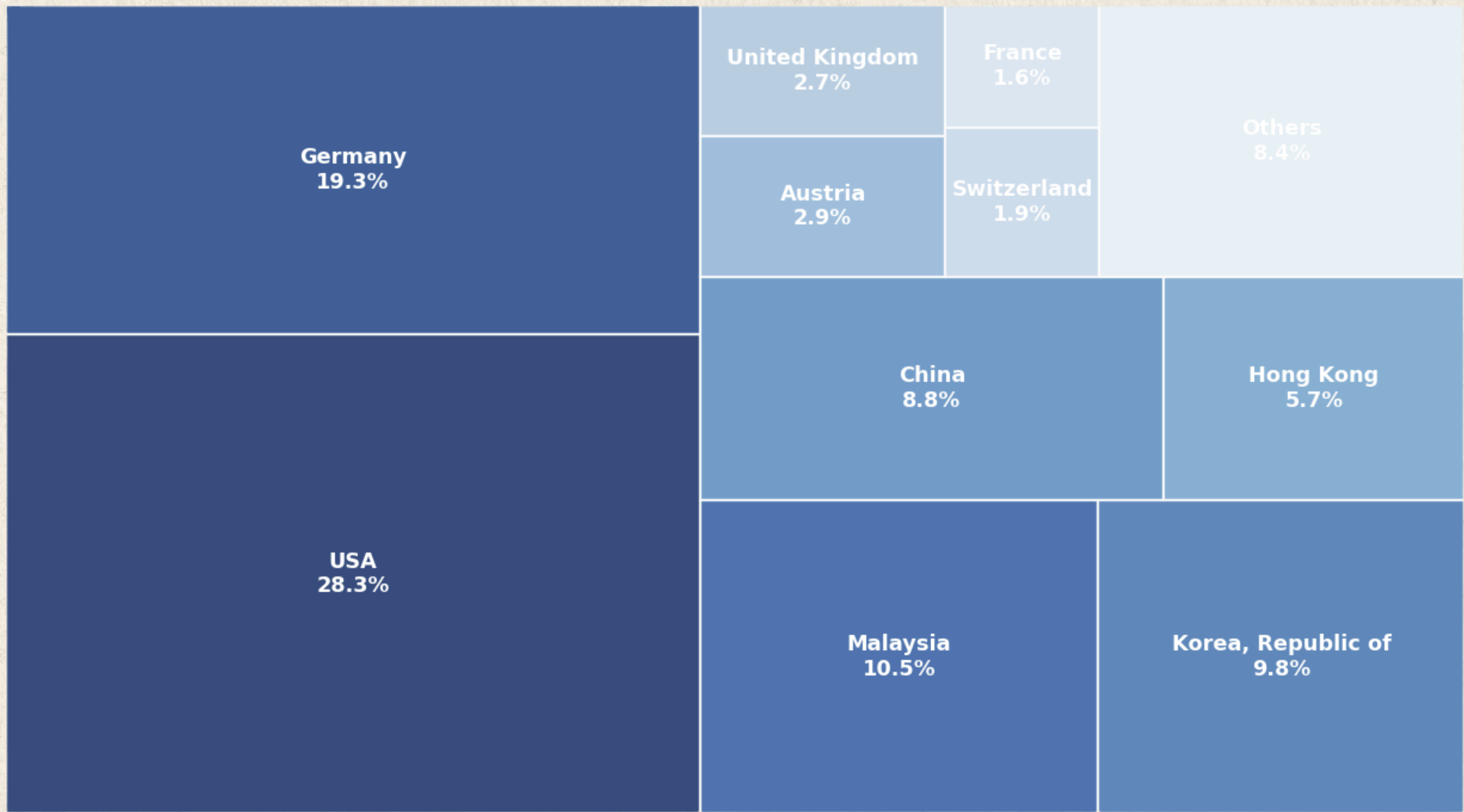
Top 5 Net Exporters vs Net Importers of Semiconductors — 2024



- Japan & Netherlands lead global exports
- China & South Korea among top importers
- Trade efficiency masks concentration risk

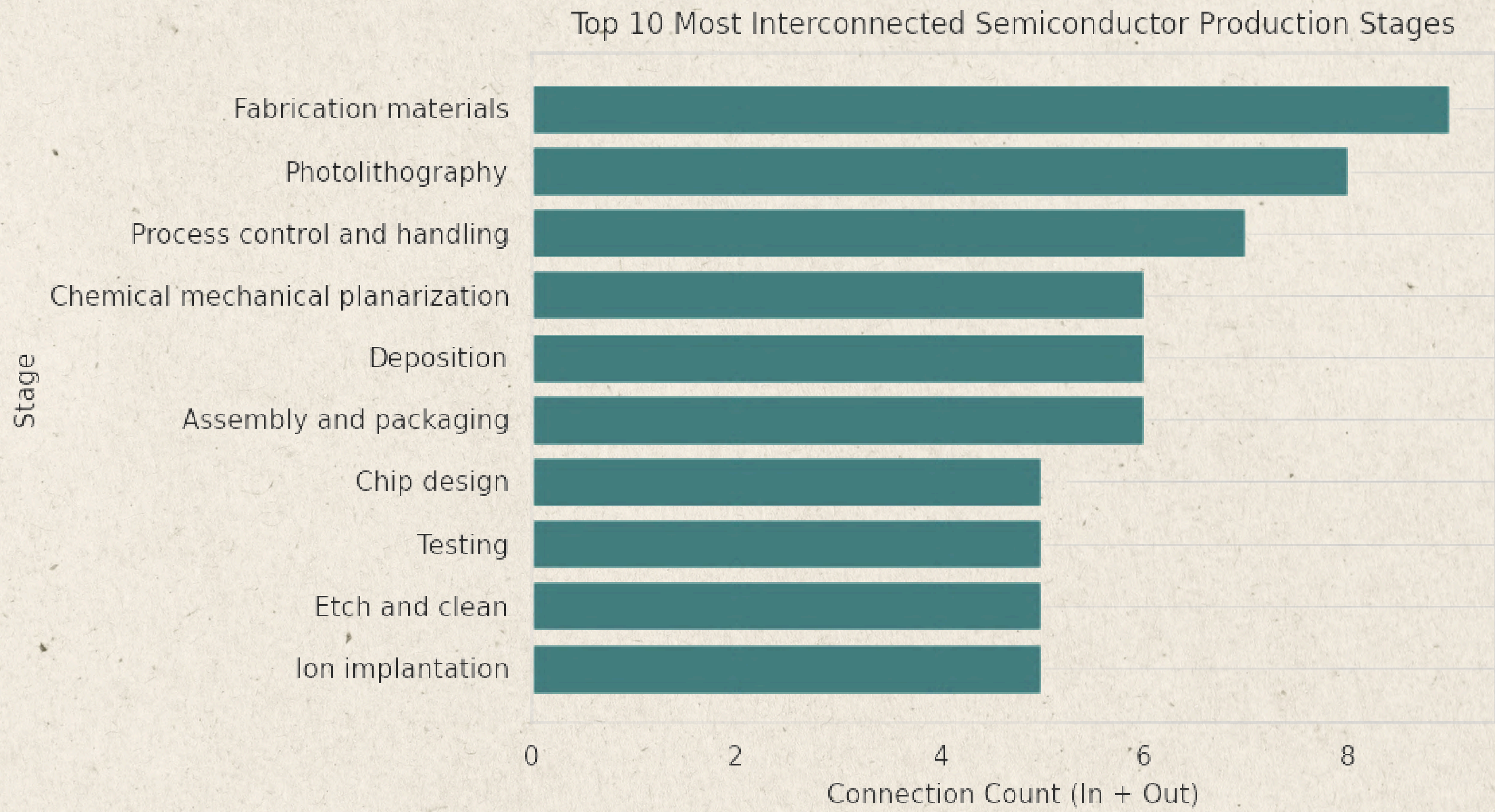
2. Market Concentration and Structural Bottlenecks

Global Semiconductor Exports by Country

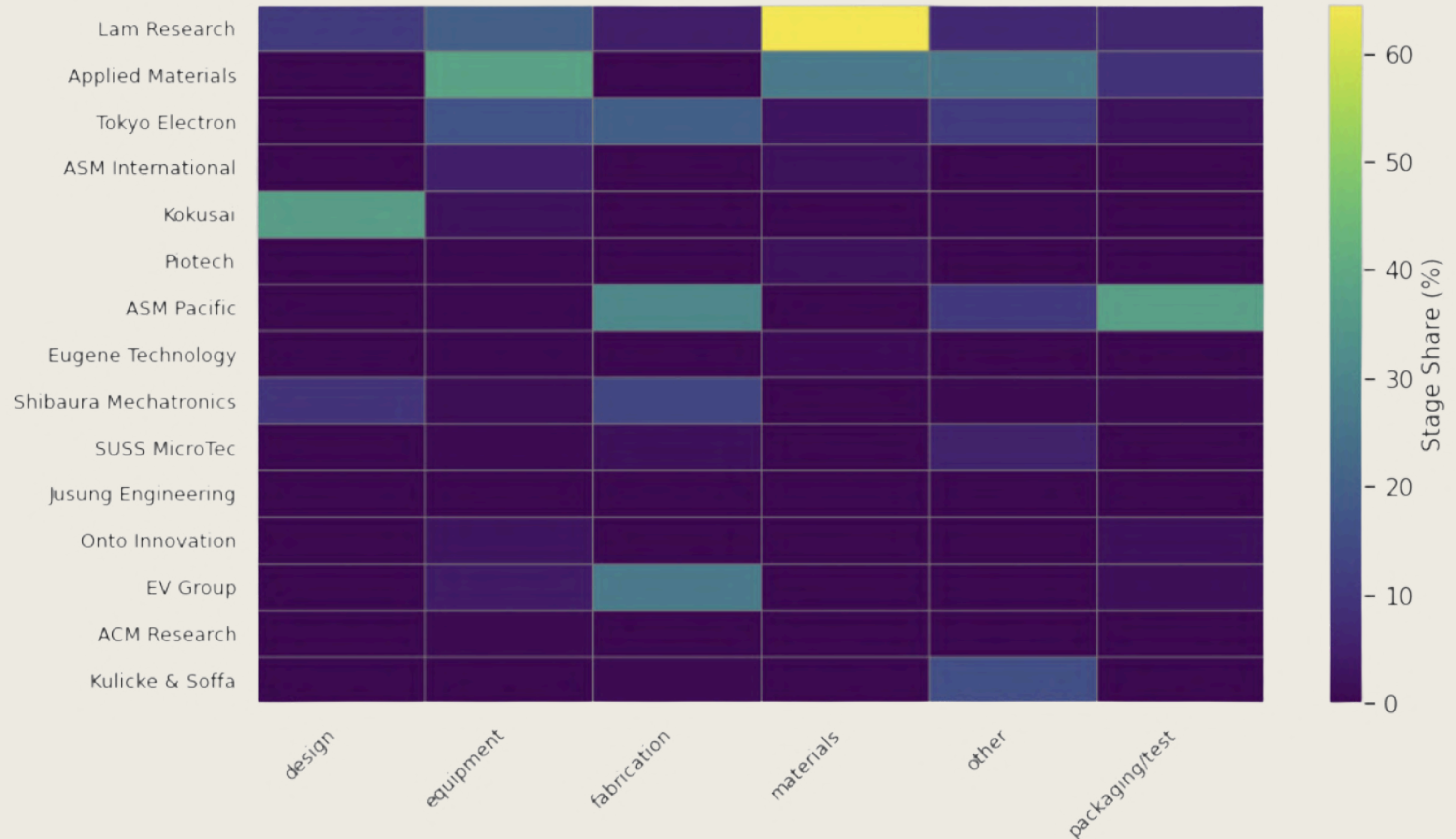


3. Network Influence and Strategic Control

Top 10 Most Interconnected Semiconductor Production Stages



Provider Concentration by Stage



4. Vulnerability, Dependency, and Macroeconomic Signals

Indicators Corealtion

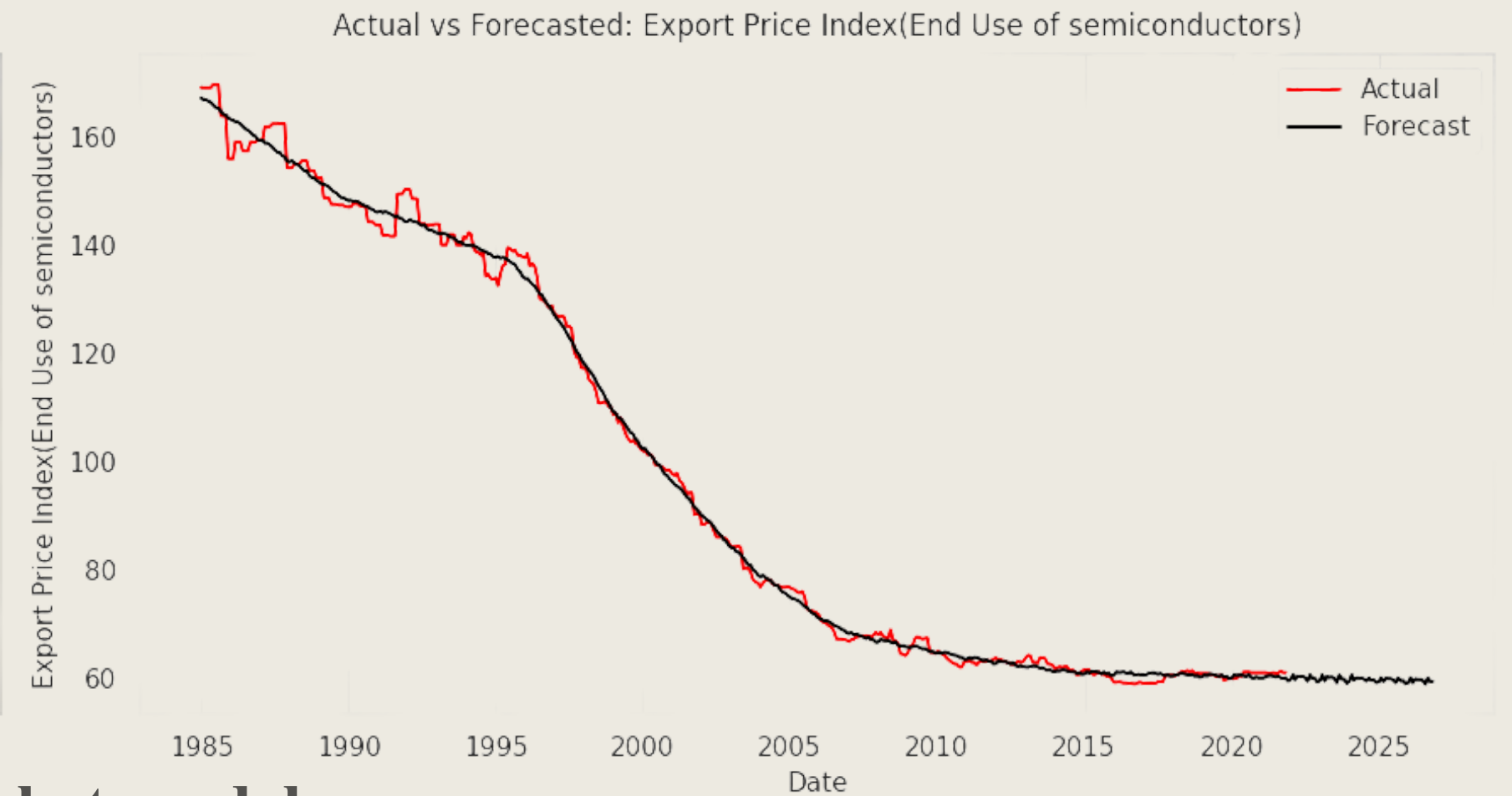
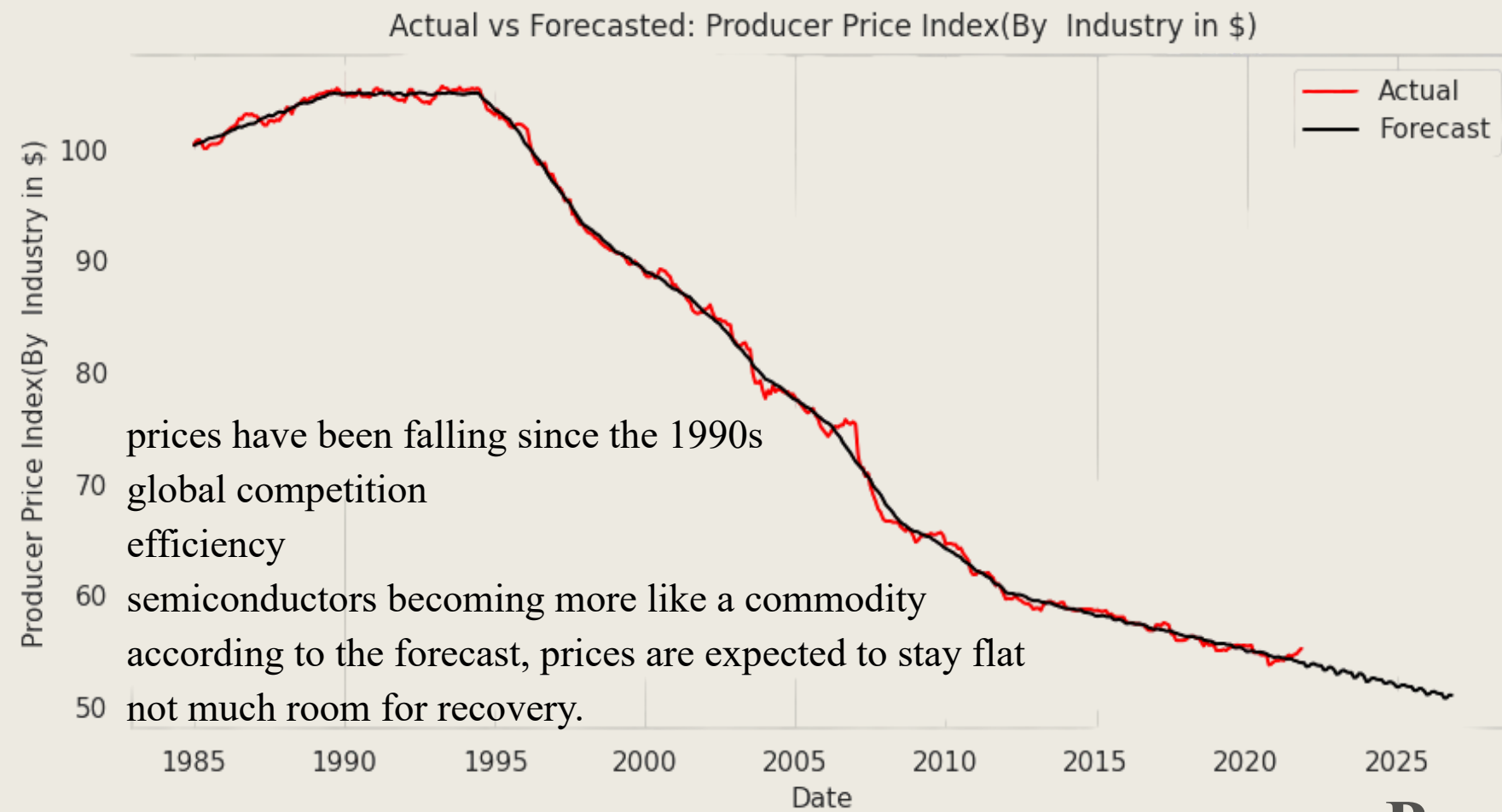
Strong positive correlations:

Producer Price Index, Export Import Price, Employment, are all highly correlated.
Sectors are highly synchronized and therefore vulnerable—if one indicator is shocked, the others will move in the same direction.

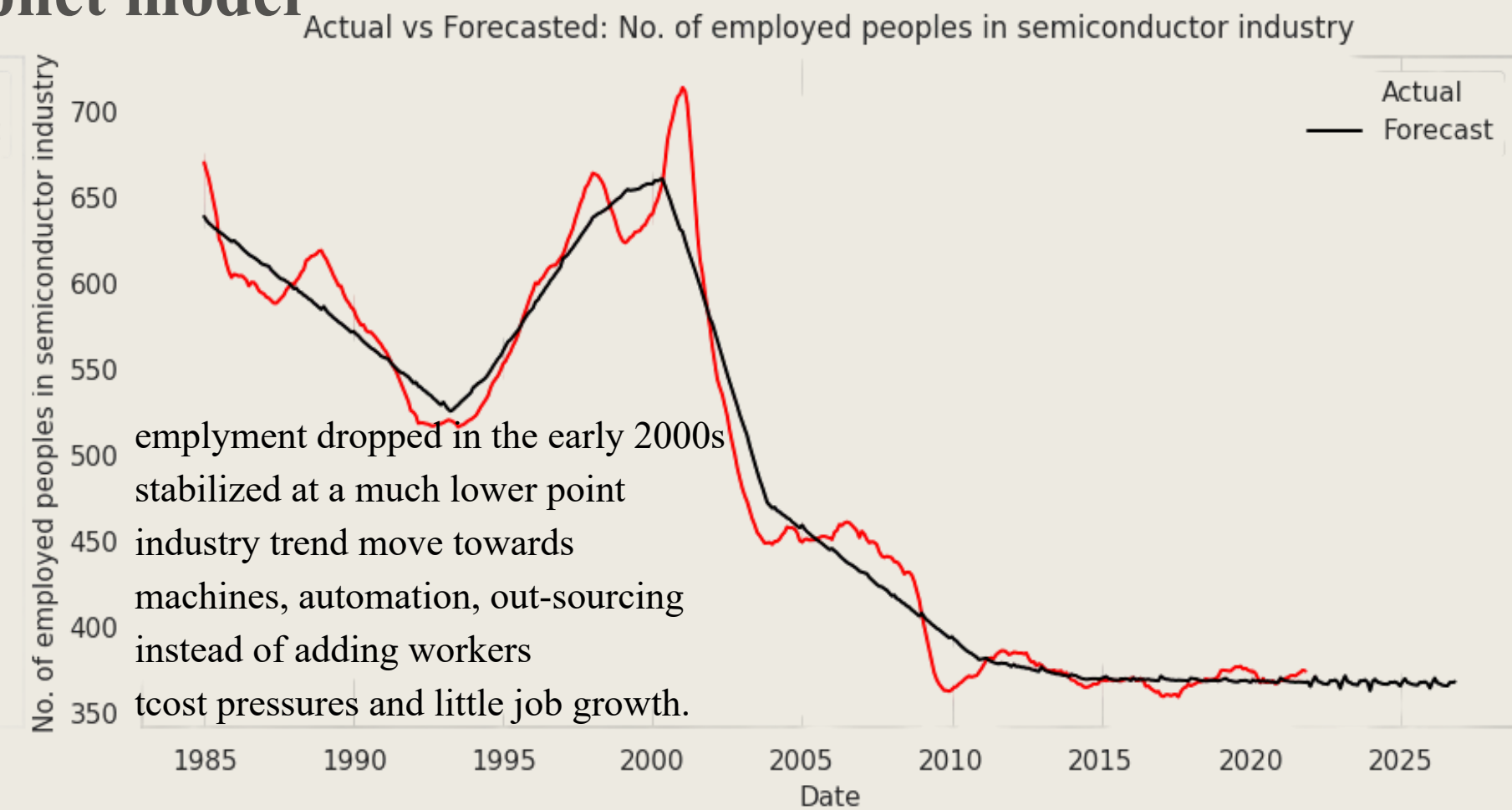
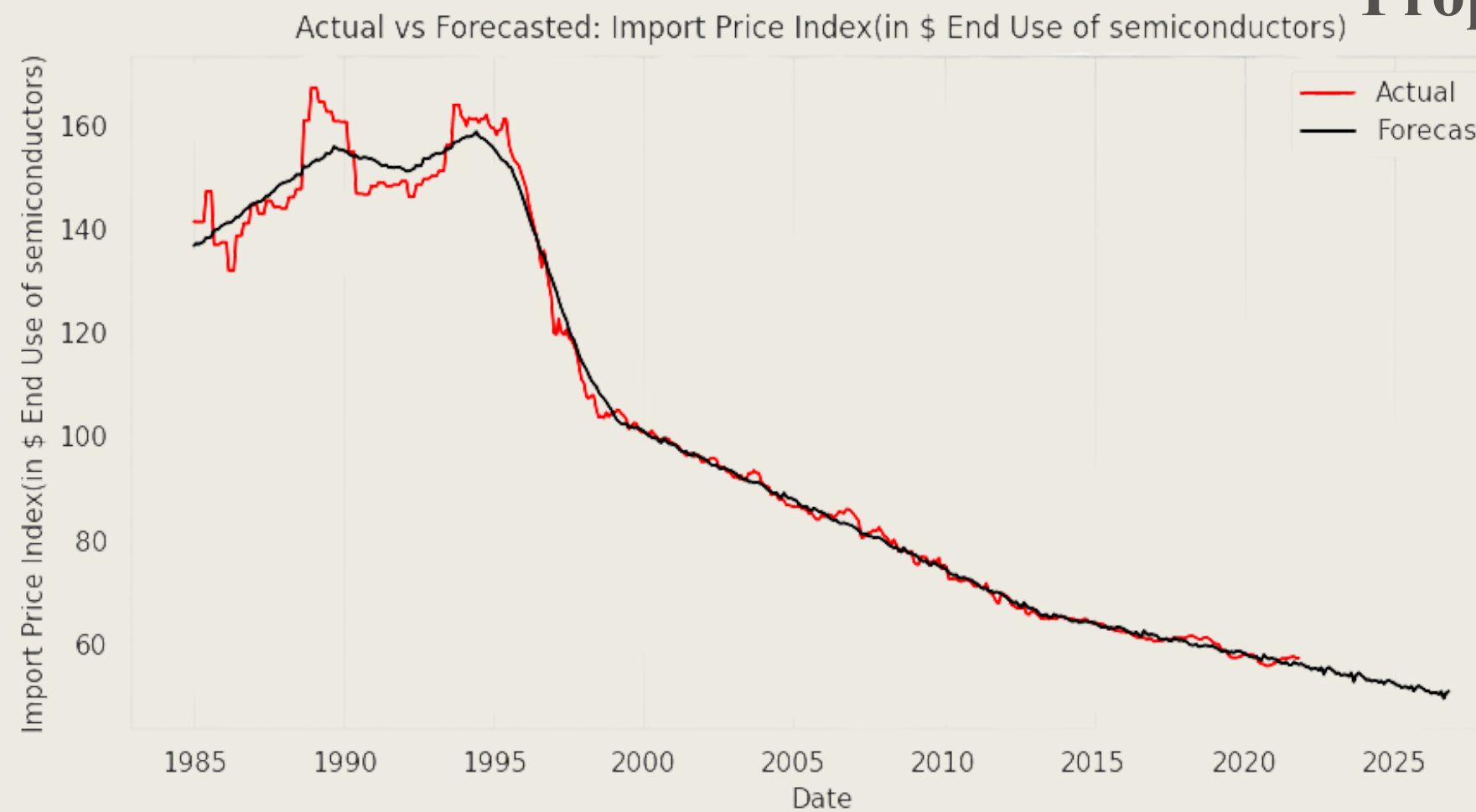
Strong negative correlations:

Import and Export Capital Goods show strong negative correlations with prices and employment.
This highlights the industry’s dependency on capital goods trade, when imports or exports are disrupted, prices and employment are strongly affected.





Prophet model



Conclusion

- Two countries and four firms dominate critical stages of production and equipment.
- Upstream stages are the real chokepoints, puts downstream stages at risk.
- Trade volume $\neq$ strategic control; power lies in technology and IP ownership.
- Resilience requires diversification, collaboration, and data-driven monitoring across the supply chain.



Thank You

Bruce Liu - Kush Vyas - Liyang Peng - Mika Ismayilli - Parishha Patel

ANY QUESTIONS ?